

Benedikt J. Koch

PHD CANDIDATE IN STATISTICS, HARVARD UNIVERSITY

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Summary

PhD candidate in Statistics at Harvard University with a focus on causal inference, counterfactual decision theory, and policy learning. Degrees from the University of Oxford and ETH Zurich. Industry research experience at Netflix in treatment-effect extrapolation and semiparametric inference with high-dimensional model embeddings. Full-time data science experience at QuantCo.

Education

Harvard University

PHD IN STATISTICS

Cambridge, MA

Sep 2023 – Present

- Advisor: Prof. Kosuke Imai
- Research in causal inference, with emphasis on experimentation, semiparametric policy learning, and counterfactual decision theory

University of Oxford

MSC IN STATISTICAL SCIENCE - GPA: 82.2/100

Oxford, GBR

Oct 2022 – Sep 2023

ETH Zurich

MSC IN APPLIED MATHEMATICS - GPA: 5.92/6.0

Zurich, CHE

Sep 2019 – Dec 2021

University of Vienna

BSC IN MATHEMATICS - GPA: 1.3/1.0

Vienna, AUT

Oct 2016 – Jul 2019

Publications

Published Papers

David Itkin, **Benedikt Koch**, Martin Larsson, and Josef Teichmann, 2022.

Ergodic robust maximization of asymptotic growth with stochastic factor processes. *Finance and Stochastics* (forthcoming).

Working Papers

Benedikt Koch, Kosuke Imai, and Tomasz Strzalecki, 2026.

An Axiomatic Foundation for Decisions with Counterfactual Utility.

Benedikt Koch and Kosuke Imai, 2025.

Statistical Decision Theory with Counterfactual Loss.

Work Experience

Netflix

MACHINE LEARNING & INFERENCE RESEARCH CONTRACTOR

Boston, MA

Sep 2025 – May 2026

- Developed a fast, debiased, low-rank CATE extrapolation framework to personalize promotions for new titles
- Built a semiparametric, nuclear-norm-regularized model to pool weak signals across high-dimensional model embeddings
- Designed scalable estimation routines and inference diagnostics for policy learning and targeting decisions

Netflix

EXPERIMENTATION & CAUSAL INFERENCE INTERN

Los Gatos, CA

Jun 2025 – Aug 2025

- Estimated promotion lift with IV designs using randomized assignments across experiments
- Built doubly robust targeting policies for personalized promotions, validated in a live A/B test
- Prototyped low-rank assortment methods for large-scale recommendation optimization

QuantCo

DATA SCIENTIST

Berlin, DEU

Sep 2021 – Sep 2022

- Designed an ML-based health-insurance pricing model serving over 800,000 policyholders
- Built actuarial decision tools supporting allocation of €120 million in funds
- Re-engineered the pricing pipeline, reducing runtime by 100x and improving reproducibility

HSBC

DATA SCIENCE & ENGINEERING INTERN

London, GBR
Jun 2021 – Aug 2021

- Developed an improved credit risk-scoring model for lending decisions
- Designed a data aggregation method that increased usable training data by 12x
- Built scalable data transformations and analysis pipelines in Apache Spark

Ministry of Defense

MATHEMATICAL ANALYST

Vienna, AUT
Feb 2019

- Developed a trajectory prediction algorithm for defense analytics applications

Aon Austria

DATA ANALYST

Vienna, AUT
Aug 2018 – Sep 2018

- Analyzed insurance data to support actuarial modeling and risk assessment

Austrian Armed Forces

PROSPECTIVE OFFICER

AUT
Sep 2015 – Present

- Military service to become an officer of the Austrian Reserve Forces (EF)
- Trained and instructed groups of up to 120 conscripts

Teaching Experience

Harvard University

TEACHING FELLOW

Cambridge, MA

- STAT 210: Probability I
- STAT 234: Sequential Decision Making
- STAT 286: Causal Inference with Applications

ETH Zurich

STUDENT ASSISTANT

Zurich, CHE

- Probability Theory and Statistics